Background Dynamics, Geodesic Perturbations, and Joint Observational Constraints

From the Exact $n = 3$ ΛCDM Limit to CMB, RSD, and Lensing Tests

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Abstract

We present the consolidated observational paper of the Grid-Clock Cosmology (GCC) programme. The minimum homogeneous grid sector interpolates between a clustering matter-like phase M and a smooth vacuum-like phase V through a transition scale factor a_t and shape parameter n. Its conserved density is

$$\rho_G(a) = \rho_{G0}\left[\frac{1+(a_t/a)^n}{1+a_t^n}\right]^{3/n},$$

and the internal exchange is

$$Q = (3-n)H\frac{\rho_M\rho_V}{\rho_G}.$$

At $n = 3$, $Q = 0$ and the model is exactly equivalent to spatially flat ΛCDM. We combine the earlier low-redshift validation with a covariant geodesic perturbation closure, $Q_M^\mu = Qu_M^\mu$, a CLASS implementation, Pantheon+ supernovae with full statistical and systematic covariance, low-redshift and BOSS BAO, BOSS DR12 redshift-space distortions, compressed primary-CMB constraints, and the official Planck 2018 lensing-reconstruction likelihood.

The low-redshift-only diagonal-supernova analysis produced an intermediate optimum at $n = 2.7114$, but that value is not the final series estimate. After the perturbations are treated consistently and CMB, full-covariance supernovae, direct velocity-based RSD, and lensing are included, the sampled profile minimum lies near $n = 2.98$. The exact correspondence point is higher by only $\Delta\chi^2 = 0.076$ before the official lensing likelihood and by only $\Delta\chi^2 = 0.219$ after it. The earlier value $n = 2.7114$ is disfavoured by approximately $\Delta\chi^2 = 8.3$ in the lensing-augmented fixed profile. Information criteria favour the simpler $n = 3$ limit.

The minimum isotropic geodesic GCC model is therefore compatible with current observations but provides no evidence for nonzero homogeneous M--V exchange. Variable effective grid number, direction dependence, and cyclic evolution are identified as separate open extensions and are not included in the reported likelihood.

Keywords: *discrete cosmology; emergent spacetime; interacting vacuum; ΛCDM; CMB; CMB lensing; redshift-space distortions; CLASS; Pantheon+; BOSS.*

1 Introduction and Relation to Paper I

Paper I defined GCC as a discrete-substrate framework and separated foundational postulates from effective cosmological closures. The present paper asks a narrower question: can the minimum homogeneous M--V sector and its geodesic perturbations satisfy distance, growth, CMB, and lensing constraints while retaining an exact ΛCDM null point?

The consolidated presentation resolves an apparent tension between the former Papers II and III. The value $n = 2.7114$ arose from a preliminary low-redshift analysis using diagonal supernova errors and a minimum growth closure. It is retained here as an intermediate validation result. The geodesic CLASS analysis with full supernova covariance, CMB information, direct RSD, and official Planck lensing supersedes it for the final scientific conclusion.

The analysis follows a strict nested-model protocol. First, the analytic $n = 3$ mapping must reproduce ΛCDM background. Second, the perturbation implementation must reproduce CMB spectra, lensing spectra, matter power, and growth observables at the same point. Only after these null tests are passed is a departure in n interpreted as a physical exchange.

2 Minimum Homogeneous GCC Sector

2.1 Unified grid-sector equation of state

The minimum homogeneous GCC sector is specified by an effective equation-of-state function

$$w_G(a) = -\frac{1}{1+(a_t/a)^n}, \tag{2.1}$$

where a is the cosmological scale factor normalized to $a = 1$ today, $a_t > 0$ is the internal transition scale factor, and $n > 0$ controls the sharpness and exchange structure of the transition.

At early times, $a \ll a_t$, one has

$$w_G(a) \to 0, \tag{2.2}$$

so the unified sector behaves as pressureless matter. At late times, $a \gg a_t$,

$$w_G(a) \to -1, \tag{2.3}$$

so it approaches a vacuum-like state. At $a = a_t$, the two internal sectors are equal and

$$w_G(a_t) = -\frac{1}{2}. \tag{2.4}$$

The corresponding internal-transition redshift is

$$z_t = \frac{1}{a_t} - 1. \tag{2.5}$$

The transition is an internal equality of M and V contributions; it is not, in general, identical to the redshift at which the total cosmic expansion begins to accelerate.

2.2 Energy-density evolution

Conservation of the total GCC stress-energy tensor in a homogeneous Friedmann background gives

$$\frac{d\rho_G}{d\ln a} = -3[1 + w_G(a)]\rho_G. \tag{2.6}$$

Substitution of Equation (2.1) yields the exact solution

$$\rho_G(a) = \rho_{G0}\left[\frac{1+a_t^n a^{-n}}{1+a_t^n}\right]^{3/n}. \tag{2.7}$$

In redshift variables, $a = (1+z)^{-1}$,

$$\rho_G(z) = \rho_{G0}\left[\frac{1+a_t^n(1+z)^n}{1+a_t^n}\right]^{3/n}. \tag{2.8}$$

For a spatially flat universe containing radiation, baryons, and the GCC sector, the dimensionless expansion function is

$$E_G^2(z) = \frac{H_G^2(z)}{H_0^2} = \Omega_r(1+z)^4 + \Omega_b(1+z)^3 + \Omega_G\left[\frac{1+a_t^n(1+z)^n}{1+a_t^n}\right]^{3/n}, \tag{2.9}$$

with flatness condition

$$\Omega_r + \Omega_b + \Omega_G = 1. \tag{2.10}$$

The low-redshift analysis sets $\Omega_r = 0$ because all fitted observables lie far below radiation domination. The baryon density is fixed provisionally to $\Omega_b = 0.05$. Sensitivity to this choice is part of the robustness programme described in Section 12.

2.3 M/V decomposition

Define

$$x(a) = \left(\frac{a_t}{a}\right)^n. \tag{2.11}$$

The GCC density can be split algebraically into matter-like and vacuum-like components,

$$\rho_{M,G} = \rho_G\frac{x}{1+x}, \qquad \rho_{V,G} = \rho_G\frac{1}{1+x}, \tag{2.12}$$

so that

$$\rho_G = \rho_{M,G} + \rho_{V,G}, \qquad p_{M,G} = 0, \qquad p_{V,G} = -\rho_{V,G}. \tag{2.13}$$

The decomposition is not the introduction of two independent dark fluids. It is a representation of a single GCC component in terms of two effective phases. The observable question is whether the relative weights and exchange implied by n differ measurably from the noninteracting ΛCDM limit.

2.4 Effective exchange term

Writing the internal continuity equations as

$$\dot{\rho}_{M,G} + 3H\rho_{M,G} = Q, \tag{2.14}$$

$$\dot{\rho}_{V,G} = -Q, \tag{2.15}$$

one obtains

$$Q = (3-n)H\frac{\rho_{M,G}\rho_{V,G}}{\rho_G}. \tag{2.16}$$

Therefore

$$n = 3 \quad \Rightarrow \quad Q = 0. \tag{2.17}$$

With the sign convention above, $n < 3$ corresponds to positive Q and $n > 3$ to negative Q. The dimensionless exchange strength is

$$\frac{Q}{H\rho_G} = (3-n)\frac{\rho_{M,G}}{\rho_G}\frac{\rho_{V,G}}{\rho_G}. \tag{2.18}$$

Its maximum absolute value occurs when the two sectors are equal,

$$\left|\frac{Q}{H\rho_G}\right|_{a=a_t} = \frac{|3-n|}{4}. \tag{2.19}$$

2.5 Exact flat-ΛCDM correspondence

For $n = 3$, Equation (2.3) reduces to

$$\rho_G(a) = \rho_{G0}\frac{1+a_t^3a^{-3}}{1+a_t^3}. \tag{2.20}$$

Define

$$\Omega_\Lambda = \frac{\Omega_G}{1+a_t^3}, \qquad \Omega_c = \frac{\Omega_G a_t^3}{1+a_t^3}. \tag{2.21}$$

Then

$$E_G^2(z) = \Omega_r(1+z)^4 + (\Omega_b + \Omega_c)(1+z)^3 + \Omega_\Lambda, \tag{2.22}$$

which is exactly ΛCDM background. The equivalent matter density is

$$\Omega_m^{equiv} = \Omega_b + \Omega_G\frac{a_t^3}{1+a_t^3}. \tag{2.23}$$

This exact nesting is the central control test of the numerical analysis. It is verified not only for $E(z)$ and distance measures but also for the minimum-closure growth equation and the complete BOSS nine-dimensional likelihood.

3 Covariant Geodesic M--V Perturbation Closure

A background exchange term alone does not define a CMB model. The perturbation sector requires a covariant energy-momentum transfer prescription. The minimum closure adopted here is

$$\nabla_\mu T_M^{\mu\nu} = Q^\nu, \qquad \nabla_\mu T_V^{\mu\nu} = -Q^\nu, \qquad Q^\nu = Q\,u_M^\nu \tag{3.1}$$

$$T_M^{\mu\nu} = \rho_M u_M^\mu u_M^\nu, \qquad T_V^{\mu\nu} = -V g^{\mu\nu}, \qquad V = \rho_V \tag{3.2}$$

The energy transfer is parallel to the M-sector four-velocity. Consequently the M sector has no fifth force in its rest frame and follows geodesic motion. In the M-comoving synchronous gauge, the closure imposes

$$\theta_M = 0, \qquad \delta V = 0, \qquad \delta Q = 0 \tag{3.3}$$

The single independent GCC scalar perturbation is then

$$\delta_M' = -\frac{h'}{2} - \mathcal{H}\left[\frac{3-n}{1+(a_t/a)^n}\right]\delta_M \tag{3.4}$$

Here a prime denotes a conformal-time derivative and $\mathscr{H}$ = aH. This equation reduces exactly to the standard cold-dark-matter synchronous-gauge equation at n = 3. Unlike the ordinary CDM case, h = -2δ_M is not true when Γ is nonzero; the CLASS implementation therefore evolves the synchronous metric variable h explicitly when GCC is active.

4 CLASS Implementation and Exact $n = 3$ Null Tests

The background and perturbation closure were implemented in CLASS. The code was parameterized for CMB applications by the early-time M-sector physical density omega_M,early rather than by the present-day split, so that the CMB acoustic scale and early matter density remain well-conditioned as n varies.

The implementation was checked in two stages. First, the analytic background was tested by mapping an ordinary ΛCDM model to n = 3 and verifying machine-precision equality of H(z). Second, the scalar perturbation system was run through CLASS and compared against the ordinary CDM model. The n = 3 GCC run reproduced ΛCDM background, CMB spectra, lensing spectra, matter power spectra, and σ_8 values to numerical precision.

For RSD diagnostics, additional transfer outputs were added for the gauge-invariant cold-plus-baryon density contrast and velocity divergence. These diagnostics were used only to define the RSD observable; they do not change the Einstein equations or the CMB calculation.

5 Data Sets, Observables, and Likelihoods

5.1 Data summary

Table 1. Low-redshift observational inputs.

Probe	Data points	Redshift range	Observable	Covariance treatment
Pantheon+SH0ES	1590	z > 0.01 to ~2.26	corrected SN magnitude	diagonal uncertainty; analytic offset
Cosmic chronometers	30	0.07-1.965	H(z)	diagonal uncertainty
6dFGS BAO	1	0.106	r_d / D_V	Gaussian approximation
SDSS MGS BAO	1	0.15	D_V r_d,fid / r_d	Gaussian approximation
BOSS DR12 consensus	9	0.38, 0.51, 0.61	D_M, H, $f\sigma_8$	full 9x9 covariance
Total	1631			

The published Pantheon+ compilation contains 1701 light curves [2]. The $z > 0.01$ cut and finite-error selection leave 1590 measurements in the present implementation. The supernova covariance limitation is important and is treated explicitly in Section 12.

5.2 Supernova distances

For a flat background, the comoving distance is

$$D_M(z) = \frac{c}{H_0}\int_0^z \frac{dz\prime}{E(z\prime)}. \tag{5.1}$$

The luminosity distance is

$$D_L(z) = (1+z)D_M(z). \tag{5.2}$$

The theoretical magnitude can be written as

$$m_{th}(z) = 5\log_{10}\left[(1+z)\int_0^z \frac{dz\prime}{E(z\prime)}\right] + \mathcal{M}, \tag{5.3}$$

where $\mathcal{M}$ absorbs the absolute supernova magnitude, the speed of light, and H_0. For diagonal errors σ_i, the nuisance offset has the analytic least-squares solution

$$\widehat{\mathcal{M}} = \frac{\sum_i [m_i - m_{shape}(z_i)]/\sigma_i^2}{\sum_i 1/\sigma_i^2}. \tag{5.4}$$

The resulting supernova statistic is

$$\chi^2_{SN} = \sum_i \frac{[m_i - m_{shape}(z_i) - \widehat{\mathcal{M}}]^2}{\sigma_i^2}. \tag{5.5}$$

Because $\mathcal{M}$ is free, this likelihood constrains the shape of the distance-redshift relation and does not by itself provide a calibrated measurement of H_0.

5.3 Cosmic chronometers

Cosmic chronometers use the differential relation

$$H(z) = -\frac{1}{1+z}\frac{dz}{dt} \tag{5.6}$$

to estimate the expansion rate from passively evolving galaxy populations [3,4]. The likelihood is

$$\chi^2_{CC} = \sum_i \frac{[H_i - H_0 E(z_i)]^2}{\sigma^2_{H,i}}. \tag{5.7}$$

For a fixed dimensionless expansion history $E(z)$, H_0 can be profiled analytically:

$$\widehat{H}_0 = \frac{\sum_i E_i H_i/\sigma^2_{H,i}}{\sum_i E_i^2/\sigma^2_{H,i}}. \tag{5.8}$$

This relation was also used in the accelerated dense-profile computation.

5.4 Low-redshift BAO

The isotropic BAO distance is

$$D_V(z) = [z D_H(z) D_M^2(z)]^{1/3}, \tag{5.9}$$

where

$$D_H(z) = \frac{c}{H(z)}. \tag{5.10}$$

The 6dFGS measurement is [5]

$$\frac{r_d}{D_V(0.106)} = 0.336 \pm 0.015. \tag{5.11}$$

For the SDSS MGS measurement [6], the Gaussian approximation used here is

$$D_V(0.15)\frac{r_{d,fid}}{r_d} = 664 \pm 25\ Mpc, \tag{5.12}$$

with $r_{d,fid} = 148.69\ Mpc$.

The sound horizon r_d is free. Therefore, BAO primarily constrains the product $H_0 r_d$ when the early-universe calibration is not supplied.

5.5 BOSS DR12 nine-dimensional consensus vector

The final BOSS likelihood uses the consensus vector [7]

$$\mathbf{d} = \left\{D_M \frac{r_{d,fid}}{r_d}, H\frac{r_d}{r_{d,fid}}, f\sigma_8\right\}_{z=0.38,0.51,0.61}, \tag{5.13}$$

with $r_{d,fid} = 147.78\ Mpc$. The observed values are listed in Table 2.

Table 2. BOSS DR12 consensus vector.

z	Observable	Observed value	Marginal uncertainty
0.38	D_M r_d,fid / r_d [Mpc]	1528.650	22.362
0.38	H r_d / r_d,fid [km s^{-1} Mpc^{-1}]	81.2369	1.9054
0.38	fσ_8	0.502242	0.045095
0.51	D_M r_d,fid / r_d [Mpc]	2007.410	26.501
0.51	H r_d / r_d,fid [km s^{-1} Mpc^{-1}]	88.3372	1.9420
0.51	fσ_8	0.459128	0.037721
0.61	D_M r_d,fid / r_d [Mpc]	2274.010	31.893
0.61	H r_d / r_d,fid [km s^{-1} Mpc^{-1}]	95.5946	2.0928
0.61	fσ_8	0.419163	0.034435

The correlated statistic is

$$\chi^2_{BOSS} = (\mathbf{t}-\mathbf{d})^T \mathbf{C}^{-1} (\mathbf{t}-\mathbf{d}), \tag{5.14}$$

where $\mathbf{C}$ is the published 9×9 covariance matrix. The matrix was symmetrized numerically and verified to be positive definite before Cholesky evaluation.

The earlier BAO-only and RSD-only BOSS calculations were diagnostic steps only. They are not added to Equation (5.14), because doing so would double-count the same galaxy sample.

5.6 Total likelihood and model parameters

The final statistic is

$$\chi^2_{tot} = \chi^2_{SN} + \chi^2_{CC} + \chi^2_{6dF+MGS} + \chi^2_{BOSS,9D}. \tag{5.15}$$

The fitted parameter sets are

$$\theta_{\Lambda CDM} = \{H_0, \Omega_m, r_d, \sigma_8, \mathcal{M}\}, \tag{5.16}$$

$$\theta_{GCC} = \{H_0, a_t, n, r_d, \sigma_8, \mathcal{M}\}. \tag{5.17}$$

The baryon fraction is fixed to $\Omega_b = 0.05$ in the GCC run. The supernova offset $\mathcal{M}$ is profiled analytically but is counted as a fitted parameter in model selection.

5.7 AIC and BIC

For Gaussian likelihoods, the information criteria are [10,11]

$$AIC = \chi^2_{\min} + 2k, \tag{5.18}$$

$$BIC = \chi^2_{\min} + k\ln N, \tag{5.19}$$

where k is the number of fitted parameters and $N = 1631$ is the total number of data entries. Smaller values are preferred. In the present comparison, GCC has one more parameter than ΛCDM.

5.8 CMB and lensing likelihoods

The observational analysis is preliminary but multi-probe. It combines a compressed Planck primary-CMB likelihood, Pantheon+ supernova distances using the full statistical plus systematic covariance matrix, low-redshift BAO, BOSS DR12 consensus distances and RSD, and the official Planck 2018 lensing reconstruction likelihood. The full official Planck high-l and low-l polarization likelihoods were not run in this environment, so the results should not be described as a final Planck posterior.

Table 3. Observational inputs for the joint CMB, distance, growth, and lensing profile.

Probe	Data used	Treatment in this paper
Primary CMB	compressed Planck 2018 TT/TE/EE with approximate low-l TT	profiled CMB likelihood; exact CLASS spectra reevaluated at fixed-n optima
Supernovae	Pantheon+SH0ES, zHD > 0.01, 1590 entries	full STAT+SYS covariance; magnitude offset profiled analytically
Low-z BAO	6dFGS and SDSS MGS	Gaussian distance constraints
BOSS DR12	D_M, H, $f\sigma_8$ at z = 0.38, 0.51, 0.61	final 9D covariance; direct velocity-based GCC RSD
CMB lensing	Planck 2018 lensing reconstruction	official native likelihood, 9 bands, 8 <= L <= 400

6 Low-Redshift Validation as an Intermediate Test

6.1 Integration and optimization

Distances were evaluated by cumulative numerical integration of $1/E(z)$ on a fixed redshift grid. The minimum-closure growth equation was integrated in $x = \ln a$ from $a_i = 0.01$ to $a = 1$ with matter-era initial conditions

$$D(a_i) = a_i, \qquad D'(a_i) = a_i. \tag{6.1}$$

The first-stage fits used global differential evolution followed by bounded local optimization. The final BOSS nine-dimensional analysis used multi-start derivative-free Powell minimization, which converged to the same minimum from several starting points.

A numerical difficulty appeared when a dense profile in n was first attempted with finite-difference L-BFGS-B optimization: nuisance parameters remained at their starting values over extended intervals and jumped discontinuously between branches. That profile was rejected. The validated profile was produced after analytically profiling H_0 and σ_8, reducing the remaining search to the expensive variables a_t and $q = H_0 r_d$. A Numba-compiled fixed-step growth integrator reduced the mean time per fixed-n point to approximately 0.021 seconds. A 301-point profile therefore completed in 6.47 seconds.

6.2 Exact background correspondence test

For arbitrary test values of Ω_b, Ω_c, and Ω_Λ, the mapping

$$a_t^3 = \frac{\Omega_c}{\Omega_\Lambda} \tag{6.2}$$

was used to compare GCC at $n = 3$ with ΛCDM. Over $0 \leq z \leq 5$, the maximum relative difference in the expansion function was

$$\max_z \left|\frac{E_G - E_{\Lambda CDM}}{E_{\Lambda CDM}}\right| = 4.33 \times 10^{-16}. \tag{6.3}$$

The corresponding luminosity-distance error was at machine precision. This verifies both the analytic correspondence and its numerical implementation.

6.3 Exact growth correspondence test

Using the equivalent matter density from Equation (2.8), the $n = 3$ GCC growth solution was compared with ΛCDM solution. The maximum relative differences were

$$\max \left|\frac{D_G - D_\Lambda}{D_\Lambda}\right| = 1.35 \times 10^{-15}, \tag{6.4}$$

$$\max \left|\frac{(fD)_G - (fD)_\Lambda}{(fD)_\Lambda}\right| = 1.39 \times 10^{-15}. \tag{6.5}$$

Thus the nested equivalence extends to the minimum-closure linear-growth calculation.

6.4 Full-likelihood correspondence test

The final $n = 3$ GCC fit gives

$$H_0 = 67.0035978, \qquad a_t = 0.7674604, \tag{6.6}$$

$$r_d = 146.6398064\ Mpc, \qquad \sigma_8 = 0.7830417, \tag{6.7}$$

$$\chi^2 = 723.008486. \tag{6.8}$$

These values agree with ΛCDM fit to the displayed precision. The agreement simultaneously checks the background mapping, distance calculations, growth solver, covariance convention, and nuisance-parameter handling.

6.5 Sequential addition of data

The evolution of the GCC best fit is shown in Figure 1 and Table 4. Supernova distances alone favour $n \simeq 2.20$, but the likelihood improvement is small. Adding direct expansion and increasingly precise BAO information moves the optimum toward $n = 3$.

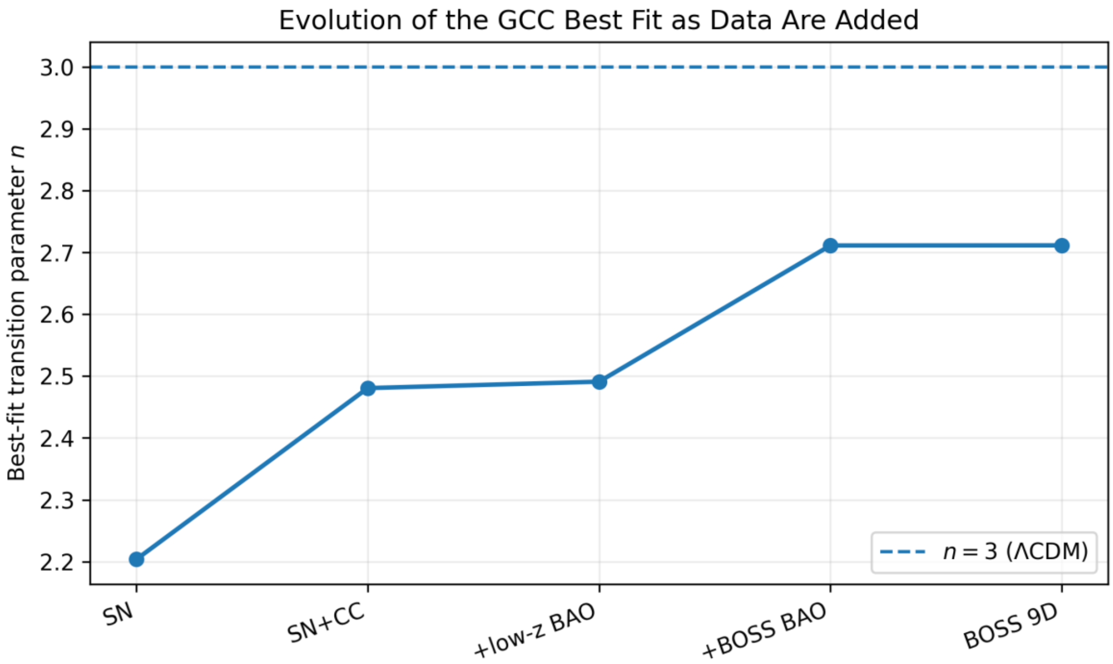

Figure 1. Evolution of the best-fit GCC transition parameter as observational probes are added. The dashed line is the exact flat-ΛCDM correspondence point.

Table 4. Sequential low-redshift model comparison. Positive ΔAIC and ΔBIC favour flat ΛCDM.

Data	n_best	χ²_Λ	χ²_G	Δχ²	ΔAIC	ΔBIC
SN	2.2037	697.505	697.076	-0.429	+1.571	+6.94
SN + CC	2.4806	712.211	711.759	-0.452	+1.548	+6.94
SN + CC + low-z BAO	2.4910	713.802	713.369	-0.433	+1.567	+6.96
+ BOSS BAO-only	2.7112	715.997	715.853	-0.144	+1.856	+7.25
Final BOSS 9D	2.7114	723.008	722.865	-0.143	+1.857	+7.25

The sequence should not be interpreted as evidence that n is converging to a measured value below three. Rather, the high-precision geometric data eliminate some of the freedom present in the supernova-only distance integral, and the remaining likelihood is broad around the nested point.

6.6 Final parameter constraints

The final low-redshift fits are summarized in Table 5.

Table 5. Final low-redshift BOSS nine-dimensional fits.

Parameter or statistic	Flat ΛCDM	GCC
H_0 [km s^{-1} Mpc^{-1}]	67.0036	67.1693
Ω_m	0.34574	--
Ω_b	--	0.05000 fixed
a_t	--	0.76321
n	3 by correspondence	2.71140
r_d [Mpc]	146.6398	146.1357
σ_8	0.78304	0.79649
χ²_SN	697.5393	697.2609
χ²_CC	14.6755	14.5918
χ²_low-z BAO	1.5902	1.6013
χ²_BOSS,9D	9.2035	9.4109
χ²_total	723.0085	722.8650
AIC	733.0085	734.8650
BIC	759.9932	767.2467

The GCC model improves the total statistic by only

$$\Delta\chi^2 = \chi^2_G - \chi^2_\Lambda = -0.1435. \qquad (6.9)$$

Because one additional parameter is fitted,

$$\Delta AIC = +1.8565, \qquad (6.10)$$

$$\Delta BIC = +7.2535. \qquad (6.11)$$

The AIC difference indicates that the models are not strongly separated by predictive fit, whereas the larger BIC penalty provides positive-to-strong support for the simpler nested description, depending on the interpretive scale adopted. The robust statement is that the present data do not justify the extra parameter n.

6.7 Profile likelihood of n

The validated dense profile is shown in Figure 2. The minimum occurs at

$$n_{best} = 2.7113 \text{ (profile interpolation)}, \qquad (6.12)$$

consistent with the direct multi-parameter optimum $n = 2.71140$.

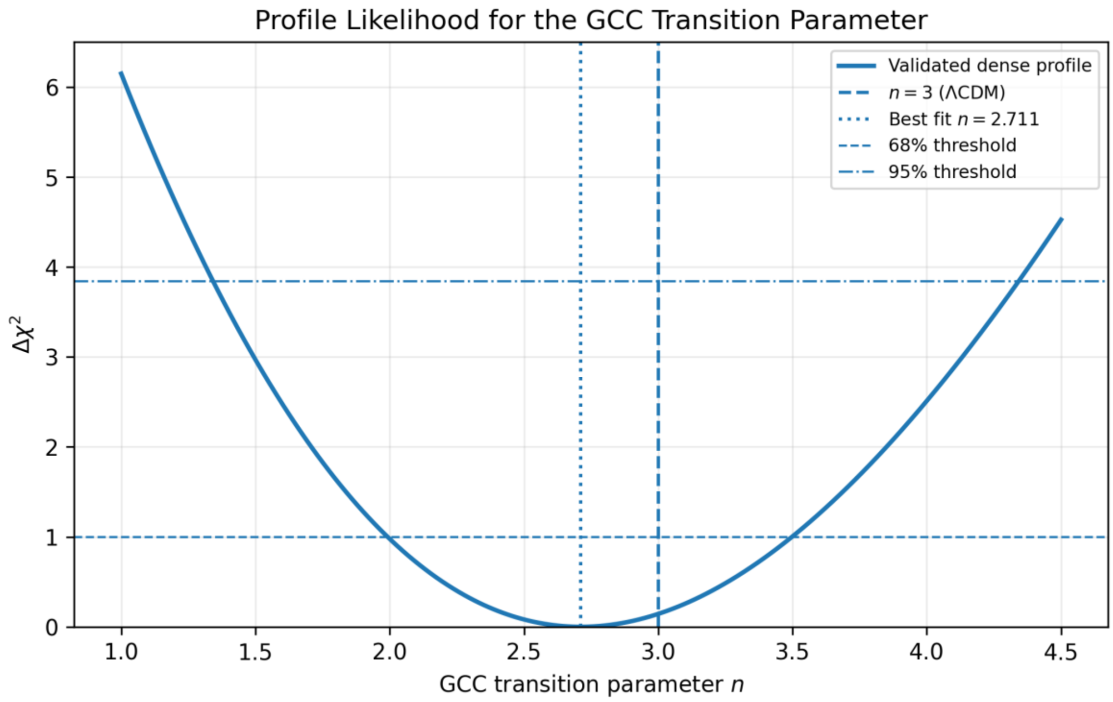

Figure 2. Validated dense profile likelihood. The horizontal lines mark the one-parameter confidence thresholds.

Cubic interpolation of the validated 301-point profile gives the approximate intervals

$$1.99 < n < 3.50 \qquad (68\%), \tag{6.13}$$

$$1.34 < n < 4.34 \qquad (95\%). \tag{6.14}$$

At the exact ΛCDM point,

$$\Delta\chi^2(n = 3) = 0.14347, \tag{6.15}$$

or, under a local Gaussian interpretation,

$$\sqrt{\Delta\chi^2} = 0.379\sigma. \tag{6.16}$$

The profile therefore provides no evidence for $n \neq 3$ and no evidence for a nonzero internal exchange.

6.8 Expansion history

The best-fit GCC and ΛCDM backgrounds differ at the sub-percent level over most of the low-redshift interval. Figure 3 shows the relative difference after both models are fitted to the full data combination.

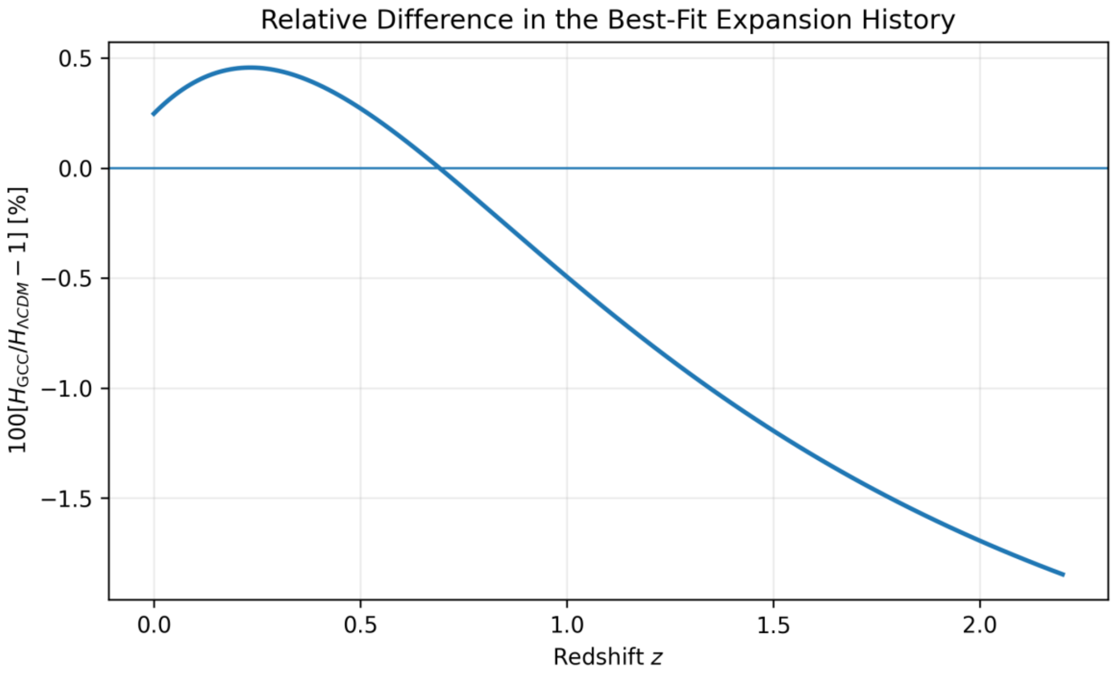

Figure 3. Relative difference between the best-fit GCC and flat-ΛCDM Hubble functions.

The small value of the difference explains why the background observables have limited power to distinguish the models. It also shows why the success of GCC is not achieved by a radically different expansion history; the best-fit solution remains close to the exact nested limit.

6.9 BOSS nine-dimensional predictions

The BOSS predictions are listed in Table 6. All marginal pulls are below approximately 1.22σ for both models.

Table 6. Final BOSS DR12 predictions.

z	Observable	Observed	ΛCDM	GCC	Pull Λ	Pull GCC
0.38	D_M r_d,f / r_d	1528.650	1541.145	1540.230	+0.559	+0.518
0.38	H r_d / r_d,f	81.2369	83.1188	83.1600	+0.988	+1.009
0.38	$f\sigma_8$	0.502242	0.471216	0.470110	-0.688	-0.713
0.51	D_M r_d,f / r_d	2007.410	1991.170	1990.310	-0.613	-0.645
0.51	H r_d / r_d,f	88.3372	90.3004	90.2233	+1.011	+0.971
0.51	$f\sigma_8$	0.459128	0.465986	0.465150	+0.182	+0.160
0.61	D_M r_d,f / r_d	2274.010	2312.740	2312.365	+1.214	+1.203
0.61	H r_d / r_d,f	95.5946	96.2827	96.0700	+0.329	+0.227
0.61	$f\sigma_8$	0.419163	0.458589	0.458283	+1.145	+1.136

The transverse, radial, and growth subsets are visualized separately in Figures 4–6.

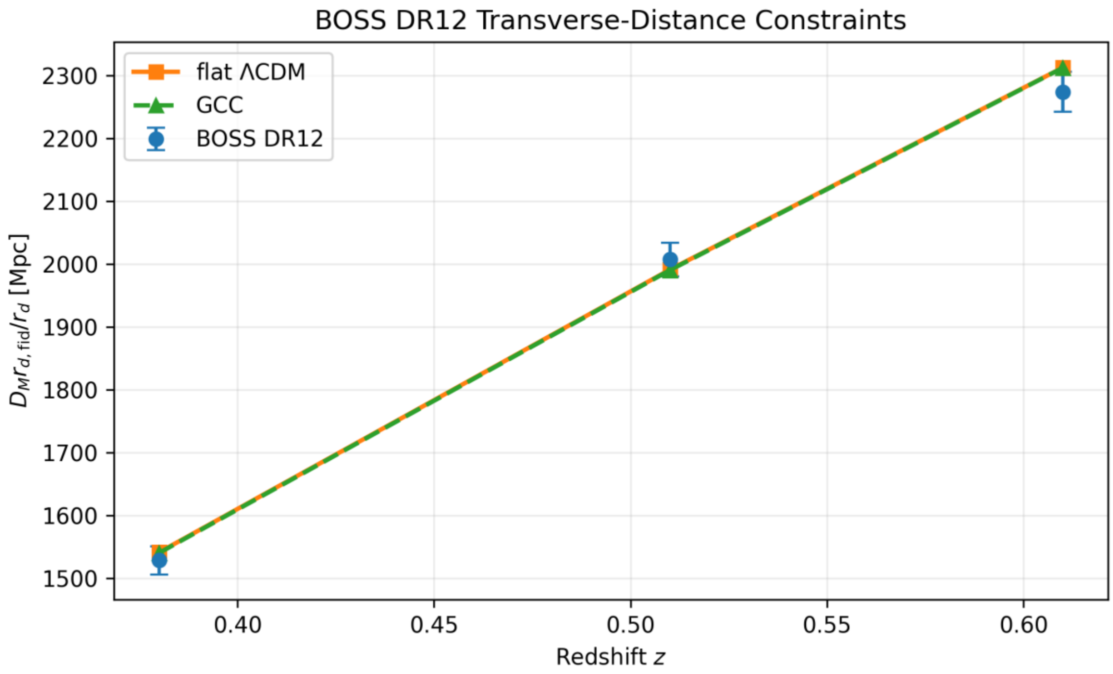

Figure 4. BOSS DR12 transverse-distance measurements and best-fit predictions.

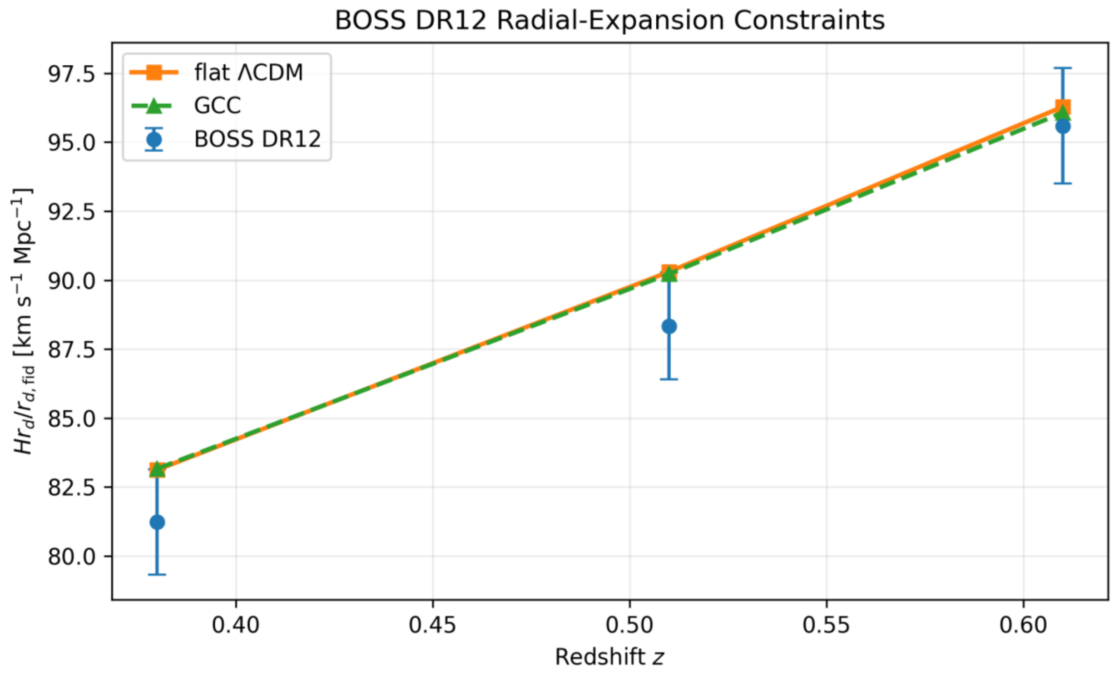

Figure 5. BOSS DR12 radial-expansion measurements and best-fit predictions.

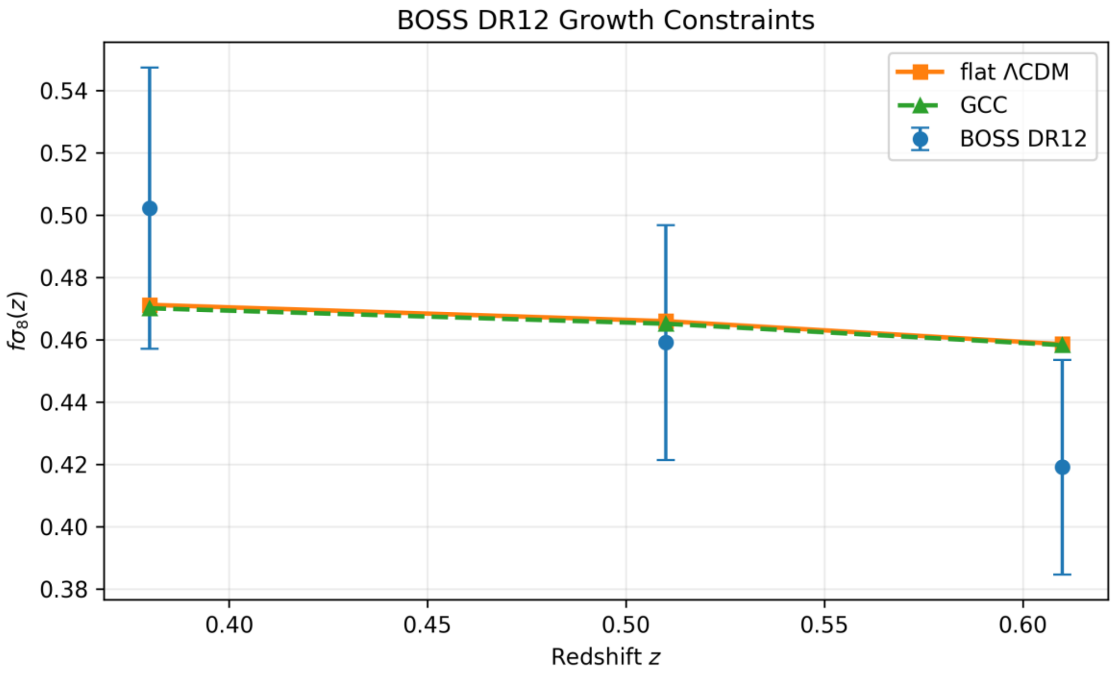

Figure 6. BOSS DR12 growth measurements and best-fit predictions.

The complete covariance, rather than the individual marginal pulls, determines the quoted BOSS χ^2. The plots nevertheless show that the two model curves are observationally almost degenerate at current precision.

6.10 Present-day sector values

For the best-fit GCC parameters,

$$a_t = 0.76321, \qquad n = 2.71140, \tag{6.17}$$

the transition redshift is

$$z_t = 0.310 \text{ (using the final 9D fit)}, \tag{6.18}$$

with small differences relative to earlier BAO-only fits caused by the updated joint likelihood.

The present GCC equation of state is approximately

$$w_G(0) = -\frac{1}{1+a_t^n} \simeq -0.68. \tag{6.19}$$

The precise value depends on the final parameter set and is not itself an equation-of-state measurement independent of the model. It describes the current mixture of the unified GCC sector.

Figure 7 shows the redshift dependence of w_G.

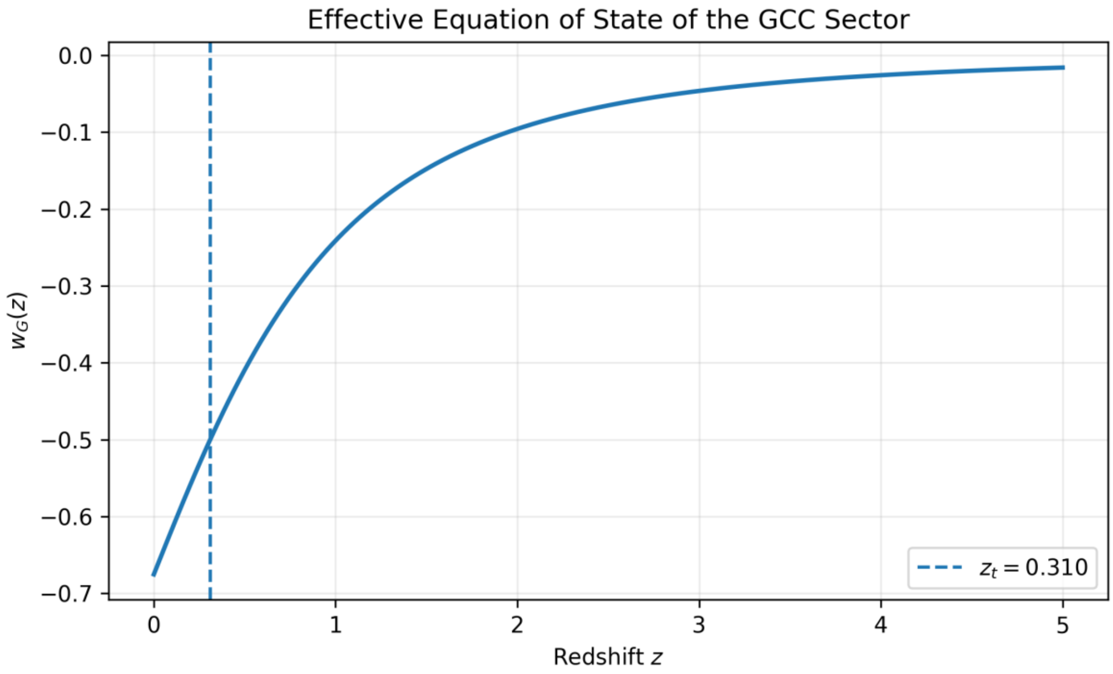

Figure 7. Effective GCC equation of state for the final best-fit transition parameters.

6.11 Exchange history

The best fit has $n < 3$, so the exchange convention of Equation (2.6) gives positive Q. The maximum dimensionless exchange is

$$\left.\frac{Q}{H\rho_G}\right|_{\max} = \frac{3-n}{4} \simeq 0.072. \tag{6.20}$$

At the present epoch, the reconstructed value is approximately

$$\frac{Q_0}{H_0\rho_{G,0}} \simeq 0.064. \tag{6.21}$$

The exchange history is shown in Figure 8.

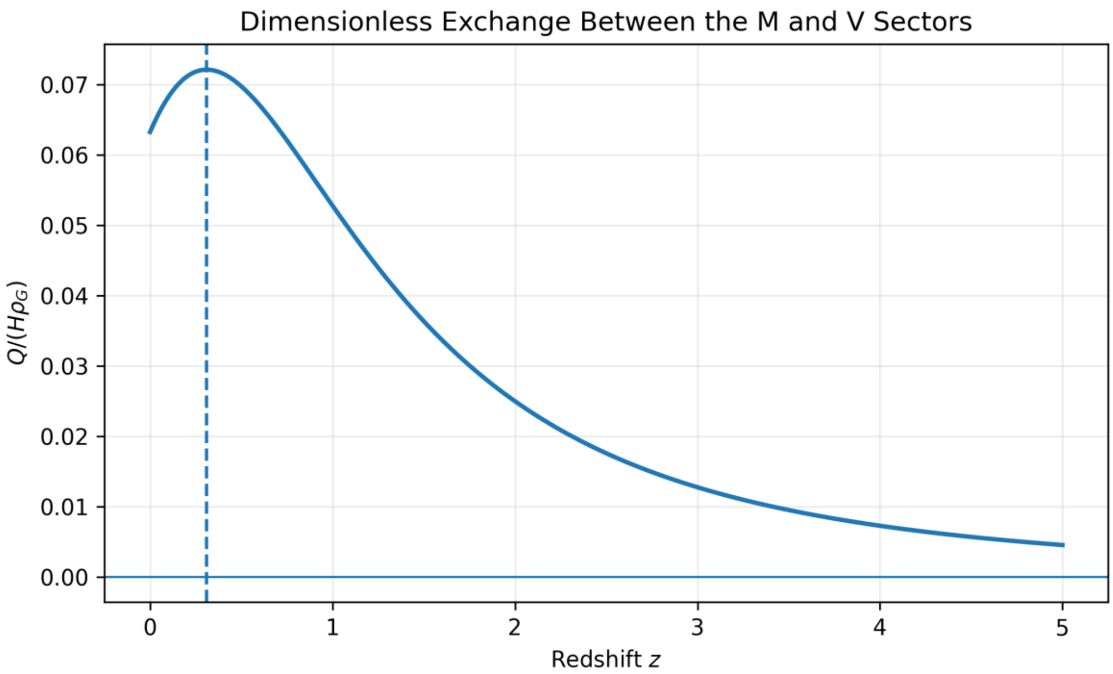

Figure 8. Dimensionless internal GCC exchange. The exchange vanishes identically at n = 3.

These numbers must not be interpreted as a detected interaction. The profile likelihood includes $Q = 0$ well inside the one-sigma region.

6.12 Sector decomposition

The effective density fractions of baryons, the GCC matter-like sector, and the GCC vacuum-like sector are shown in Figure 9.

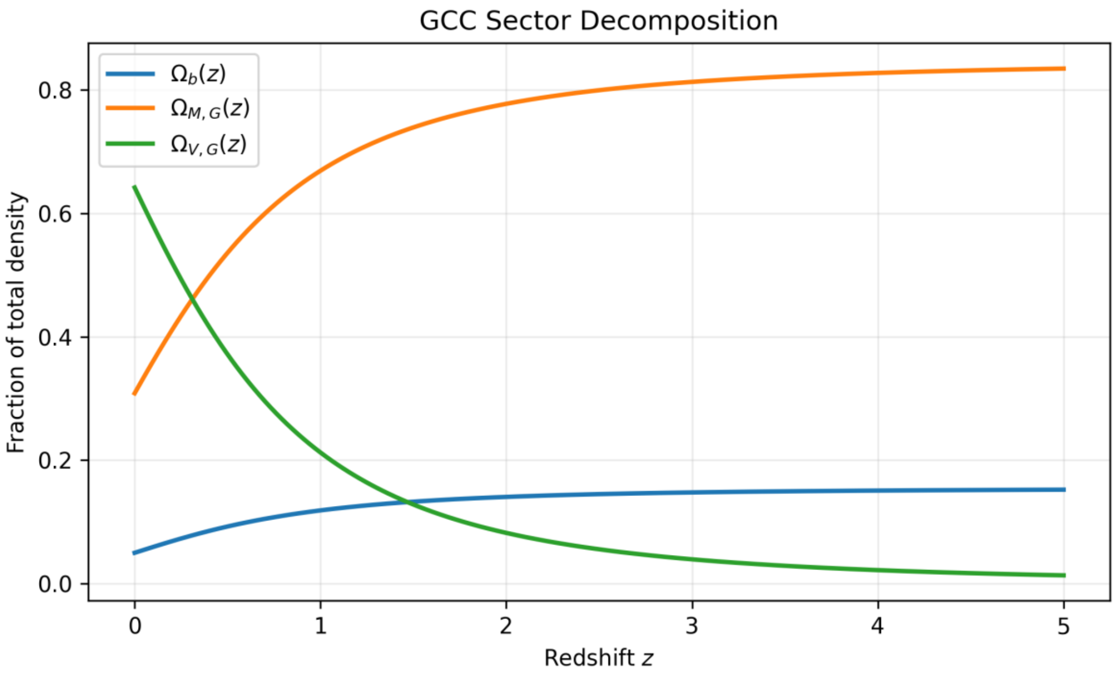

Figure 9. Decomposition of the best-fit GCC background into baryonic, matter-like grid, and vacuum-like grid fractions.

For the preceding BAO-only background fit, the present effective clustering fraction was approximately $\Omega_{cl,0} = 0.361$. The final nine-dimensional fit changes this value only modestly. The GCC fit therefore does not remove the need for an effective nonbaryonic clustering component at the phenomenological level. Rather, it replaces independent cold-dark-matter and vacuum components by two phases of a unified grid sector.

6.13 Growth-shape prediction

Before fitting σ_8, the minimum closure predicts a distinct redshift dependence of fD. Figure 10 shows the fractional difference from the separately fitted ΛCDM background.

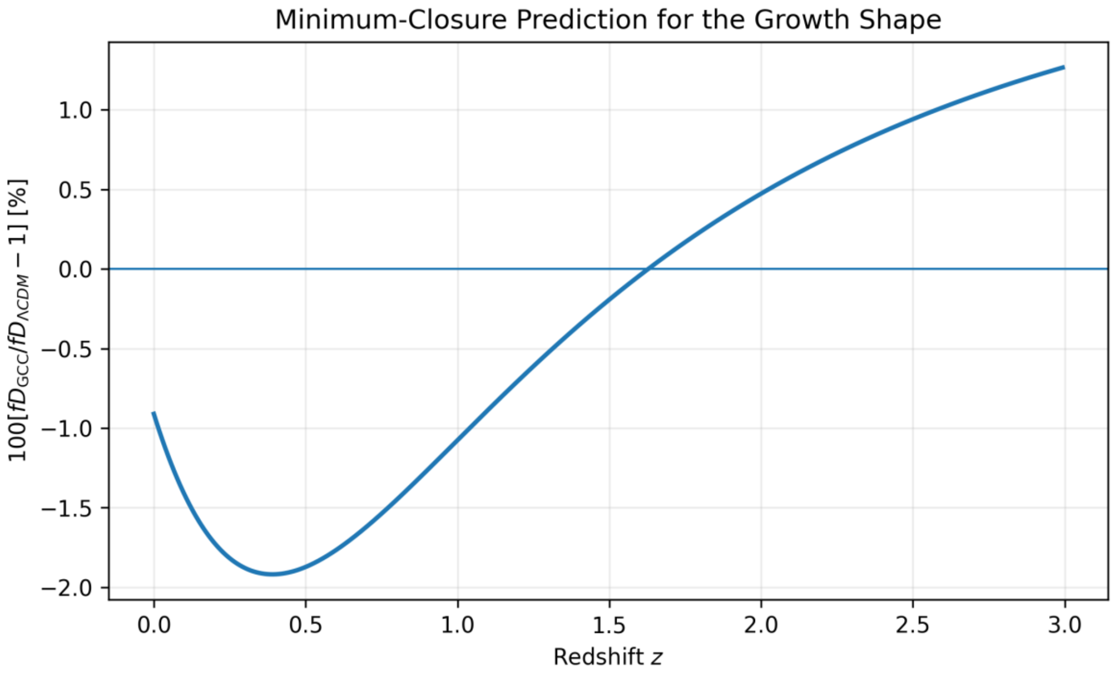

Figure 10. Minimum-closure prediction for the relative growth-shape difference.

Representative values from the frozen-background diagnostic are given in Table 7.

Table 7. Minimum-closure growth-shape comparison.

z	D_G / D_Λ	f_G / f_Λ	(fD)_G / (fD)_Λ - 1
0.0	1.0000	0.9908	-0.92%
0.1	1.0007	0.9851	-1.42%
0.3	1.0028	0.9785	-1.88%
0.5	1.0052	0.9762	-1.87%
0.8	1.0088	0.9770	-1.44%
1.0	1.0109	0.9787	-1.06%
1.5	1.0148	0.9836	-0.18%
2.0	1.0173	0.9877	+0.48%

The largest low-redshift effect is a suppression of about two percent near $z \simeq 0.3$–0.6. The sign reverses above $z \sim 1.6$. This sign-changing pattern is more informative than a single present-day amplitude.

However, the BOSS data allow the normalization to shift. The final fits give

$$\sigma_{8,0}^{\Lambda CDM} = 0.7830, \tag{6.22}$$

$$\sigma_{8,0}^{G} = 0.7965. \tag{6.23}$$

The approximately 1.7% larger GCC normalization compensates for the lower fD over the BOSS redshift interval, leaving almost identical $f\sigma_8$ predictions. Breaking this degeneracy requires an independent amplitude measurement or a broad, high-precision redshift sequence.

6.14 What the fit establishes

The analysis establishes four positive results.

First, GCC has a numerically exact flat-ΛCDM limit. This statement is stronger than a qualitative resemblance: the background function, distances, minimum-closure growth, nuisance parameters, and full joint likelihood agree at machine precision when $n = 3$ and the correspondence map is applied.

Second, the best-fit minimum GCC model is compatible with all low-redshift data sets included here. No parameter is driven to a boundary, the fitted sound horizon remains near the standard phenomenological scale, and no BOSS observable has an anomalously large residual.

Third, the model produces a definite growth-shape signature. Although current BOSS data do not resolve it, a percent-level, redshift-dependent difference exists before normalization freedom is used.

Fourth, the framework supplies a clean null test: if future data constrain $n = 3$ and the minimum closure is retained, the internal exchange vanishes and the cosmological sector is observationally equivalent to ΛCDM at the tested level.

6.15 What the fit does not establish

The analysis does not detect a nonzero exchange. The optimum $n < 3$ is less than half a Gaussian standard deviation away from $n = 3$. It is therefore not evidence for $V \to M$ conversion or for a dynamical dark sector.

The analysis does not eliminate dark phenomenology. The unified GCC component still contains an effective matter-like fraction and a vacuum-like fraction. The claim is one of unification of description, not a baryon-only universe.

The analysis does not solve the Hubble tension. The supernova magnitude offset is marginalized and r_d is free, so the inferred $H_0 \simeq 67$ is determined mainly by the chronometer/BAO combination under this likelihood construction. It is not a joint calibrated SH0ES-versus-CMB tension result.

The analysis does not provide a CMB prediction. Without a microphysical drag epoch, a perturbation-transfer four-vector, and a Boltzmann implementation, r_d remains a nuisance parameter and no TT, TE, EE, or lensing spectrum is calculated.

6.16 Information-criterion interpretation

The difference

$$\Delta AIC = +1.86 \tag{6.24}$$

is modest. By itself it would usually be interpreted as weak evidence against the extended model. The BIC difference

$$\Delta BIC = +7.25 \tag{6.25}$$

is larger because the sample contains 1631 entries, so the additional parameter is penalized more strongly. Both criteria point in the same direction: the additional transition-shape freedom has not earned its complexity.

A scientifically appropriate conclusion is therefore not that GCC is excluded, but that the minimum low-redshift data set prefers the nested $n = 3$ submodel on grounds of parsimony.

7 Direct Velocity-Based Redshift-Space Distortions

In an interacting M--V model, the logarithmic derivative of a density amplitude is not automatically the same quantity as the velocity growth measured by redshift-space distortions. We therefore define the RSD growth rate directly from the cold-plus-baryon velocity divergence:

$$f_{\mathrm{RSD}}(k, z) = -\frac{\theta_{cb}(k,z)}{aH\,\delta_{cb}(k,z)} \tag{7.1}$$

The BOSS comparison quantity is the average of f_RSD over 0.02 < k < 0.2 h Mpc^-1 multiplied by $\sigma_8\sigma_8$,cb(z). In this linear range the scale dependence of f_RSD was below about 0.1 percent for the sampled GCC models, supporting a first comparison to the compressed BOSS f σ_8 consensus vector.

Table 8. Direct velocity-based GCC RSD predictions compared with the BOSS DR12 consensus values.

n	f (z=0.38)	f (z=0.51)	f (z=0.61)
2.7114	0.4569	0.4541	0.4488
2.9000	0.4669	0.4652	0.4602
3.0000	0.4785	0.4766	0.4712
BOSS obs.	0.5022	0.4591	0.4192

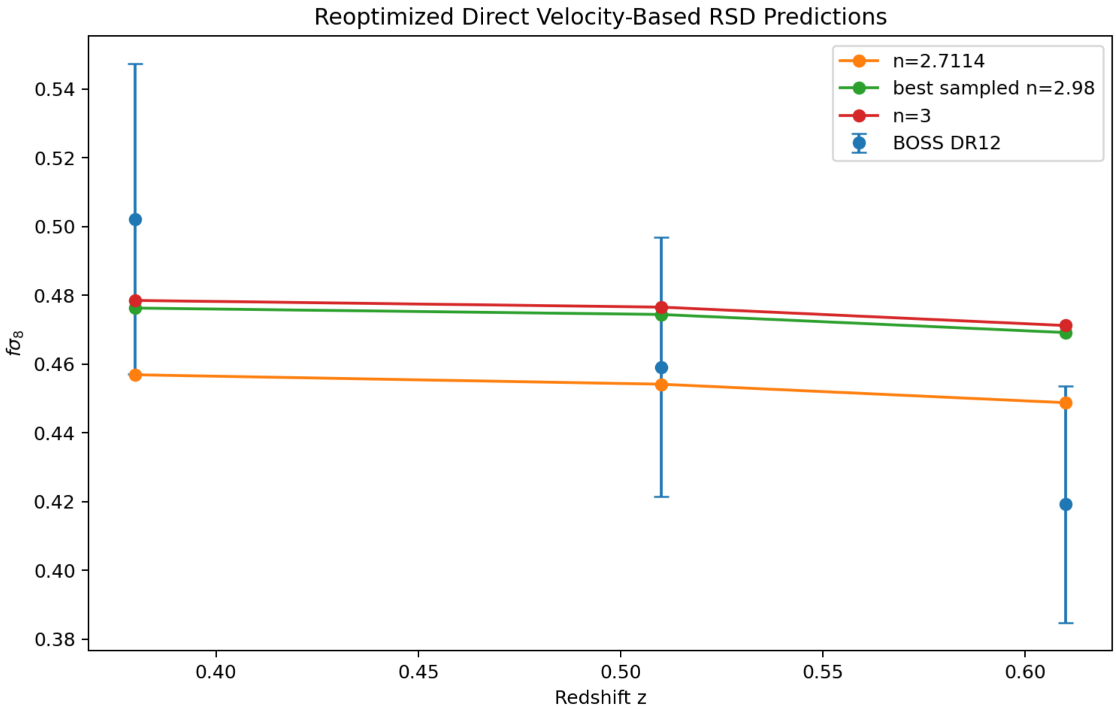

Figure 11. Reoptimized direct velocity-based RSD predictions for selected GCC models compared with the BOSS DR12 consensus values.

8 Joint Profile with Primary CMB and Full-Covariance Supernovae

We first reoptimized the joint CMB + Pantheon+ + BAO + BOSS 9D profile without the official Planck lensing reconstruction likelihood. At fixed n, the parameters omega_b, omega_M,early, H0, and ln(10^10 A_s) were numerically optimized; n_s and tau were profiled through the compressed-CMB surrogate and the final primary-CMB likelihood was reevaluated using exact CLASS spectra.

Table 9. Reoptimized CMB + full-covariance SN + BAO + BOSS 9D profile before the official Planck lensing likelihood.

n	H_0	total χ^2	$\Delta\chi^2$
2.7114	66.553	2010.302	6.688
2.8000	66.926	2006.193	2.578
2.9000	67.353	2004.820	1.205
2.9500	67.351	2003.775	0.161
2.9800	67.306	2003.614	0.000
3.0000	67.276	2003.690	0.076
3.0200	67.412	2004.081	0.467
3.0500	67.615	2004.605	0.991
3.1000	67.956	2006.154	2.539

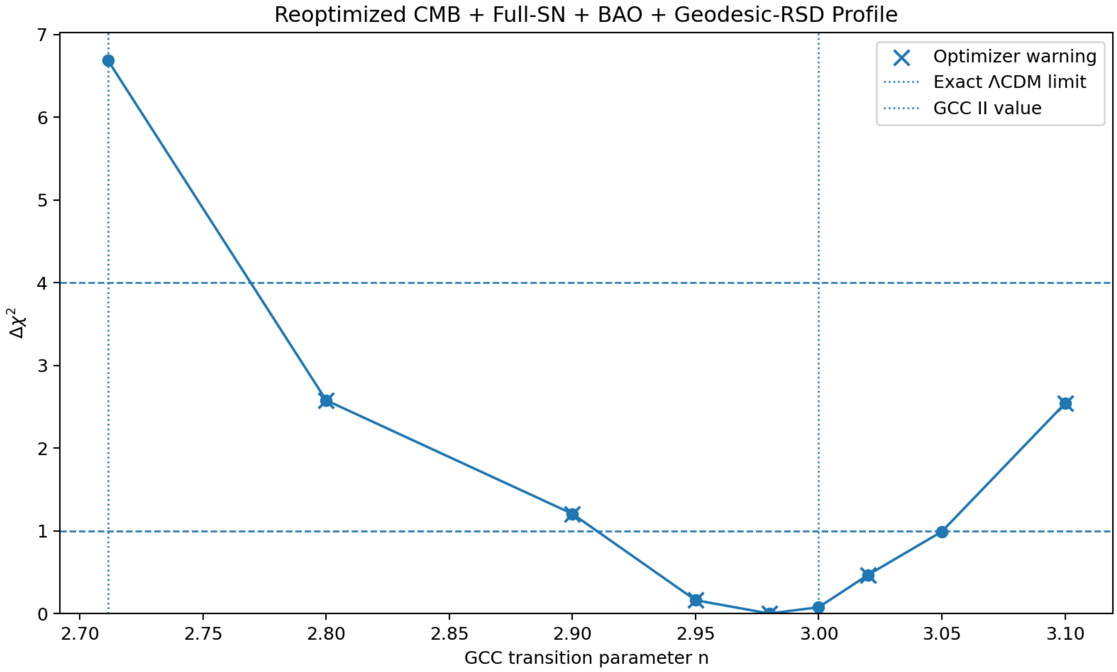

Figure 12. Reoptimized profile before the official Planck lensing reconstruction likelihood. The sampled minimum is near n = 2.98, while n = 3 is higher by only $\Delta\chi^2 = 0.076$.

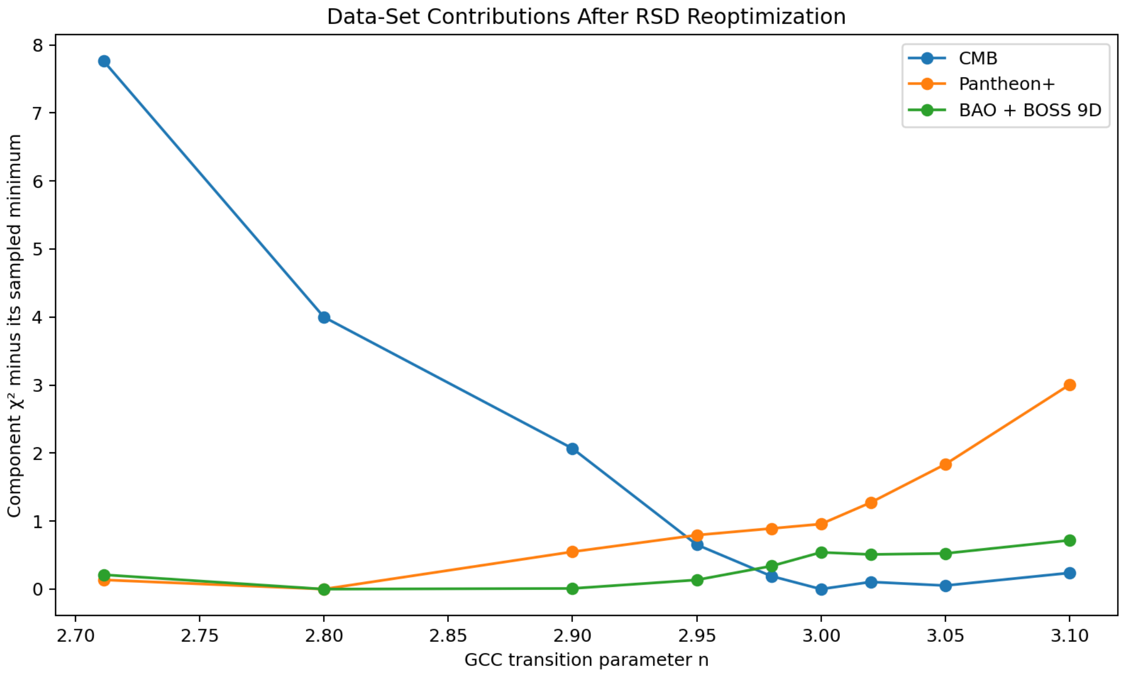

Figure 13. Contributions from the primary CMB, Pantheon+, and BAO + BOSS 9D blocks after RSD reoptimization.

The sampled minimum occurs at n = 2.98, H0 = 67.306 km s^-1 Mpc^-1, and χ^2 = 2003.614. The exact ΛCDM correspondence point n = 3 is higher by only $\Delta\chi^2$ = 0.076. This difference is far below the level required to support an additional exchange parameter. The previous earlier low-redshift value n = 2.7114 is higher by $\Delta\chi^2$ = 6.688 relative to the sampled minimum and by $\Delta\chi^2$ = 6.612 relative to n = 3.

9 Official Planck 2018 Lensing Reconstruction

The official Planck 2018 CMB-lensing reconstruction likelihood was then evaluated at the same fixed-n optimized points. This likelihood uses nine bandpowers over 8 <= L <= 400. Nonlinear corrections in

CLASS were computed with Halofit. The cosmological parameters were not reoptimized after adding the lensing likelihood, so this section is a fixed-profile addition rather than a final posterior.

Table 10. Official Planck 2018 lensing reconstruction added to the fixed-n reoptimized profile.

n	Planck lensing χ^2	total χ^2 with lensing	$\Delta\chi^2$
2.7114	10.426	2020.728	8.305
2.8000	9.781	2015.974	3.551
2.9000	9.219	2014.039	1.616
2.9500	8.744	2012.520	0.097
2.9800	8.809	2012.423	0.000
3.0000	8.952	2012.642	0.219
3.0200	8.960	2013.041	0.619
3.0500	8.973	2013.578	1.155
3.1000	9.001	2015.154	2.731

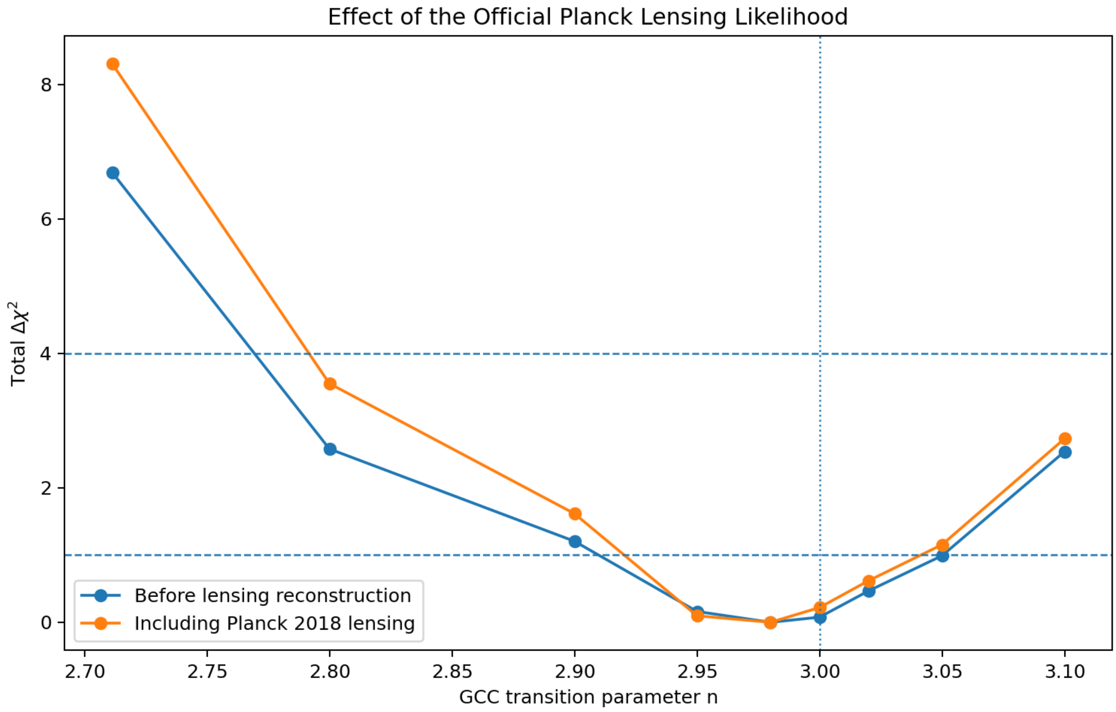

Figure 14. Total profile before and after adding the official Planck 2018 lensing reconstruction likelihood.

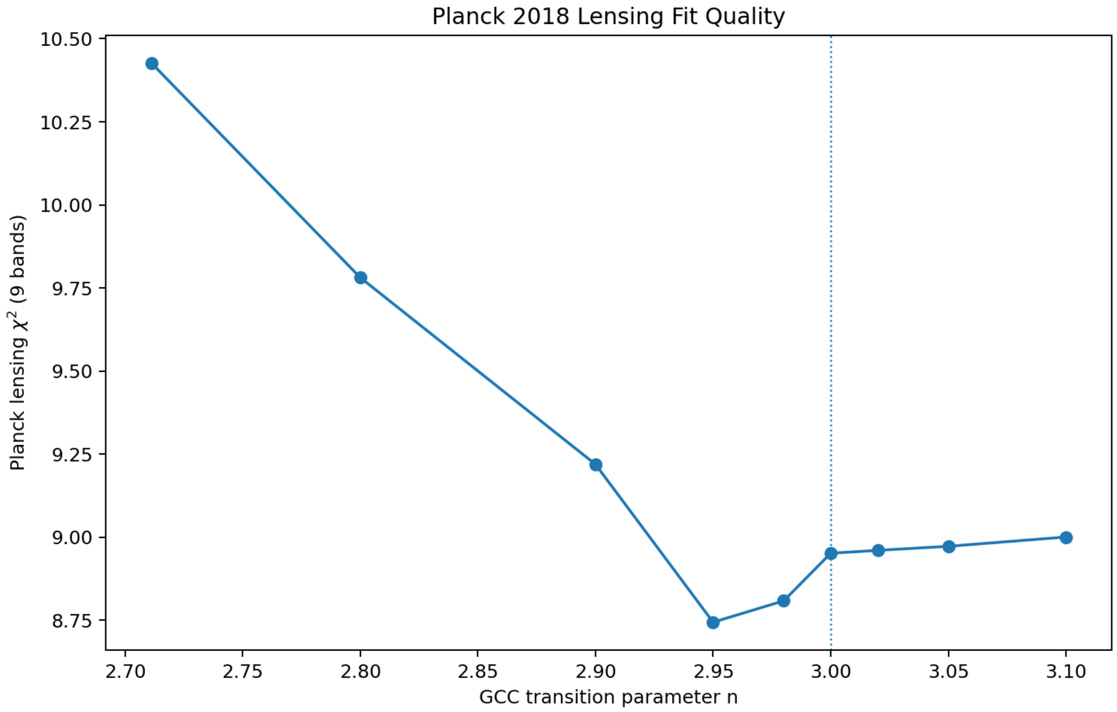

Figure 15. Absolute Planck 2018 lensing-reconstruction χ^2 values for the sampled GCC profiles.

The lensing-augmented fixed profile retains a minimum at n = 2.98, but n = 3 is only $\Delta\chi^2 = 0.219$ higher. The earlier value n = 2.7114 lies $\Delta\chi^2 = 8.305$ above the lensing-augmented sampled minimum and $\Delta\chi^2 = 8.085$ above n = 3. The lensing result therefore reinforces the conclusion that large positive M--V exchange is not favoured.

10 Joint Model Selection and Physical Interpretation

The profile improvement of n = 2.98 over n = 3 is too small to justify the additional parameter. Before lensing reconstruction, the best sampled point improved χ^2 by only 0.076 relative to n = 3, giving Delta AIC approximately +1.924 and Delta BIC approximately +7.627 for GCC relative to the exact ΛCDM correspondence point. After adding official Planck lensing, the improvement is 0.219, giving Delta AIC approximately +1.781 and Delta BIC approximately +7.488. Both information criteria therefore prefer the simpler n = 3 limit, with BIC giving the stronger penalty against the extra exchange freedom.

Physically, n < 3 corresponds to Q > 0: the M sector receives energy from the V sector. In the geodesic closure, newly generated M does not automatically carry the pre-existing density contrast. The perturbation equation therefore contains a dilution term proportional to Gamma delta_M. For values as low as n = 2.7114, the present-day Gamma/H is about 0.20, which suppresses structure growth and CMB lensing too strongly once BOSS RSD and Planck lensing are included. Near n = 2.98, Gamma/H is only of order 0.014, and the model becomes observationally almost indistinguishable from ΛCDM.

Table 11. Main interpretation of the joint profile.

Comparison	Value	Interpretation
Best sampled n before lensing	2.98	$\Delta\chi^2$= 0.076 versus n = 3; not significant
Best sampled n after lensing	2.98	$\Delta\chi^2$= 0.219 versus n = 3; not significant
earlier low-redshift value	n = 2.7114	disfavoured by $\Delta\chi^2$ ~ 8.3 after lensing
Information criteria	Delta AIC ~ +1.8, Delta BIC ~ +7.5	additional n parameter not supported
Robust conclusion	n ≃ 3	no evidence for homogeneous M--V exchange

11 Variable Effective Grid Number, Direction Dependence, and Cyclic Evolution

The likelihood analysed above assumes a fixed microscopic graph, homogeneous activation, and statistical isotropy. A broader completion may distinguish the fixed cardinality N_{grid} from an effective active count,

$$N_{\text{eff}} = N_{\text{eff}}(\tau, \mathbf{x}). \tag{11.1}$$

A minimal isotropic parameterization may be written as

$$a_{\text{phys}}(\tau, \mathbf{x}) = a_{\text{link}}(\tau, \mathbf{x}) \left[\frac{N_{\text{eff}}(\tau,\mathbf{x})}{N_{\text{eff},0}}\right]^{1/3}, \tag{11.2}$$

which contributes

$$H_{\text{phys}} = H_{\text{link}} + \frac{1}{3}\frac{\dot{N}_{\text{eff}}}{N_{\text{eff}}}. \tag{11.3}$$

This relation shows why a time-dependent effective count could be relevant to discrepancies in locally and globally inferred expansion rates. It does not establish a solution of the H_0 tension: such a claim requires a covariant stress tensor, perturbation equations, recombination-era consistency, and a new joint likelihood. Likewise, $\nabla N_{\text{eff}} \neq 0$ can source directional dependence, while a sign change in $\dot{N}_{\text{eff}}$ may supply one ingredient of cyclic evolution. These are open model classes, not post-fit explanations of the isotropic results.

A valid extension must satisfy three requirements: it must recover the $n = 3$ isotropic limit when the new field is switched off; it must use one parameter set across SN, BAO, CMB, RSD, and lensing; and its direction-dependent prediction must be tested against survey geometry and look-elsewhere effects. No-retuning is essential.

12 Limitations and Required Next Tests

The primary-CMB block used here is compressed and is not a substitute for a full official Planck high-ℓ/low-ℓ MCMC analysis with foreground nuisance parameters. The official lensing reconstruction was added along a fixed-n optimized profile rather than followed by complete reoptimization. BOSS RSD was compared through the compressed consensus vector rather than a forward model of the power-spectrum multipoles. Nonlinear bias, nonlinear velocity fields, and possible baryon--M relative bias are not derived from the grid microphysics.

The earlier low-redshift stage used diagonal Pantheon+ uncertainties, fixed baryon density, and a free sound horizon. Those choices were useful for a CMB-independent validation but cannot determine the final posterior. The reported profile intervals are diagnostic because several fixed-n optimizations produced line-search warnings.

The next decisive analysis is a public, reproducible GCC-CLASS implementation coupled to the full official Planck likelihoods, current BAO and supernova covariance products, weak-lensing and full-shape clustering data, and a formal MCMC or nested-sampling posterior. Directional and variable-N_{eff} models must be analysed as new models rather than appended after the isotropic fit.

13 Conclusions

The consolidated analysis establishes a consistent geodesic perturbation closure for the minimum GCC M--V sector and verifies the exact $n = 3$ correspondence across the background, primary CMB, CMB lensing, matter power, and growth observables. This null test is the central technical achievement of the paper.

The observational result is conservative. The earlier low-redshift optimum at $n = 2.7114$ was a useful intermediate result but is not supported by the broader data combination. The reoptimized profile lies near $n = 2.98$, while $n = 3$ is statistically indistinguishable and preferred by AIC and BIC. Present observations therefore show no evidence for homogeneous M--V exchange.

GCC remains viable at its exact standard-cosmology correspondence point. Future discriminatory power must come from sectors not already absorbed by that limit: controlled directional dependence, a covariant variable-N_{eff} extension, or the strong-curvature saturation physics developed in Paper III.

Appendix A. Derivation of the GCC Density Law

Starting from the continuity equation

$$\frac{d\rho_G}{d\ln a} = -3(1 + w_G)\rho_G \tag{A.1}$$

and Equation (2.1),

$$1 + w_G = \frac{x}{1+x}, \qquad x = \left(\frac{a_t}{a}\right)^n. \tag{A.2}$$

Since

$$\frac{dx}{d\ln a} = -nx, \tag{A.3}$$

we have

$$\frac{d\ln\rho_G}{d\ln a} = -3\frac{x}{1+x} = \frac{3}{n}\frac{1}{1+x}\frac{dx}{d\ln a}. \tag{A.4}$$

Therefore

$$d\ln\rho_G = \frac{3}{n}\frac{dx}{1+x}. \tag{A.5}$$

Integrating from $a = 1$, where $x = a_t^n$, to an arbitrary a gives

$$\ln\frac{\rho_G(a)}{\rho_{G0}} = \frac{3}{n}\ln\left[\frac{1+(a_t/a)^n}{1+a_t^n}\right], \tag{A.6}$$

which yields Equation (2.3).

Appendix B. Derivation of the Internal Exchange

From Equation (2.5),

$$\rho_{M,G} = \rho_G \frac{x}{1+x}. \tag{B.1}$$

Differentiate with respect to $\ln a$:

$$\frac{d\rho_{M,G}}{d\ln a} = \frac{x}{1+x}\frac{d\rho_G}{d\ln a} + \rho_G \frac{1}{(1+x)^2}\frac{dx}{d\ln a}. \tag{B.2}$$

Using

$$\frac{d\rho_G}{d\ln a} = -3\rho_G \frac{x}{1+x}, \qquad \frac{dx}{d\ln a} = -nx, \tag{B.3}$$

and adding $3\rho_{M,G}$, one finds

$$\frac{d\rho_{M,G}}{d\ln a} + 3\rho_{M,G} = (3-n)\rho_G \frac{x}{(1+x)^2}. \tag{B.4}$$

Because

$$\frac{\rho_{M,G}\rho_{V,G}}{\rho_G} = \rho_G \frac{x}{(1+x)^2}, \tag{B.5}$$

the cosmic-time equation is

$$\dot{\rho}_{M,G} + 3H\rho_{M,G} = (3-n)H\frac{\rho_{M,G}\rho_{V,G}}{\rho_G} = Q. \tag{B.6}$$

Appendix C. Likelihood Acceleration

The validated dense profile uses three reductions.

First, the chronometer likelihood is quadratic in H_0, so Equation (5.8) eliminates one numerical dimension.

Second, for fixed geometry and growth shape, the BOSS theory vector can be written

$$\mathbf{t} = \mathbf{t}_0 + \sigma_8 \mathbf{g}, \tag{C.1}$$

where $\mathbf{g}$ is nonzero only in the three growth entries. The full-covariance generalized least-squares solution is

$$\hat{\sigma}_8 = \frac{\mathbf{g}^T \mathbf{C}^{-1}(\mathbf{d} - \mathbf{t}_0)}{\mathbf{g}^T \mathbf{C}^{-1} \mathbf{g}}. \tag{C.2}$$

Third, the BAO geometry depends mainly on

$$q = H_0 r_d. \tag{C.3}$$

For example,

$$D_M \frac{r_{d,f}}{r_d} = \frac{c\, r_{d,f}}{q} \int_0^z \frac{dz'}{E(z')}, \tag{C.4}$$

$$H \frac{r_d}{r_{d,f}} = \frac{qE(z)}{r_{d,f}}. \tag{C.5}$$

Thus, at fixed (a_t, n), the remaining geometric optimization is one-dimensional in q. The growth equation is solved with a compiled fixed-step Runge–Kutta scheme. Together these changes reduced the final 301-point profile to 6.47 seconds on a Colab CPU.

Appendix D. Numerical Reproducibility Checklist

A reproducible implementation should verify all of the following before interpreting a GCC deviation:

1. $E_G(z; n = 3)$ agrees with ΛCDM to better than 10^{-10}.
2. D_L, D_M, and D_V satisfy the same correspondence.
3. The minimum-closure D and fD solutions agree at $n = 3$.
4. The BOSS covariance is symmetric and positive definite.
5. BOSS BAO-only and RSD-only diagnostics are not double-counted in the final nine-dimensional likelihood.
6. The dense profile has smooth nuisance-parameter trajectories with no optimizer plateaus or discontinuous branch jumps.
7. The accelerated likelihood reproduces the full fit to $|\Delta\chi^2| < 0.01$ at the optimum.
8. Parameter-boundary warnings are checked for n, a_t, r_d, and σ_8.
9. Confidence intervals are computed from a validated profile or posterior, not from an optimizer-stalled curve.
10. The supernova covariance approximation is stated explicitly.

Appendix E. CLASS Implementation Notes

In the CLASS implementation, GCC replaces the ordinary CDM and cosmological-constant dark sector when active. The M density contributes to clustering and matter transfer functions, while the V density contributes to the background pressure and density but carries no independent perturbation in the M-comoving synchronous gauge. A separate synchronous metric variable is integrated because $h = -2\delta_M$ is invalid when Γ is nonzero.

The $n = 3$ correspondence test is the central validation. Any implementation that fails to reproduce ΛCDM in this limit cannot be used for inference. The version used for this paper passed the null test for the background, CMB spectra, CMB lensing potential, linear matter power, and σ_8-related quantities.

Appendix F. Reproducibility Package

The accompanying files include the GCC-CLASS source packages, patches, profile CSV files, plots, diagnostic scripts, RSD transfer-output patch, and Planck lensing evaluation materials. The official Planck low-l EE and full high-l likelihood data must be installed separately through Cobaya or another official pipeline before a final MCMC analysis can be performed.

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